

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418558
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Ding; Jiahao et al.

Detection of malicious domains

Abstract

Disclosed are systems and methods that monitor for malicious and unauthorized behaviors, determine categories for detected malicious behaviors, determine why a domain is determined to be malicious, and provide information to users that identifies the categories and reasons as to why a domain is determined to be malicious. In some implementations, the disclosed systems and methods may be utilized to provide monitoring security to customers of a cloud service. For example, customers of a cloud service may maintain an account with the cloud service and the disclosed implementations may be utilized to protect those accounts from malicious attacks and cybercrimes such as, but not limited to, spam, phishing, malware, botnets, etc.

Inventors: Ding; Jiahao (Long Island City, NY), Dubey; Abhinandan (Jersey City, NJ), Stinson; Christopher (Columbia, MD), Bickford; Jeffrey Earl (Thornton, CO), Coskun; Baris (Glen Rock, NJ), Torkamani; MohamadAli (Scarsdale, NY)

Applicant: Amazon Technologies, Inc. (Seattle, WA)

Family ID: 1000007403793

Assignee: Amazon Technologies, Inc. (Seattle, WA)

Appl. No.: 18/478107

Filed: September 29, 2023

Publication Classification

Int. Cl.: H04L9/40 (20220101)

U.S. Cl.:

CPC H04L63/1433 (20130101); H04L63/1416 (20130101); H04L63/1425 (20130101);

Field of Classification Search

CPC: H04L (63/1433); H04L (63/1416); H04L (63/1425)

USPC: 726/22-25

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7739211	12/2009	Coffman	706/45	H04L 63/145
10091229	12/2017	Stute	N/A	H04L 63/1433
11743282	12/2022	Torkamani et al.	N/A	N/A
11886586	12/2023	Chang	N/A	G06F 21/566
2016/0065597	12/2015	Nguyen	726/22	H04L 63/1441
2017/0063912	12/2016	Muddu	N/A	G06F 3/0482
2017/0310693	12/2016	Howard	N/A	H04L 63/166
2019/0028508	12/2018	Jeng	N/A	H04L 43/16
2021/0314349	12/2020	Liu	N/A	G06N 20/00
2021/0326744	12/2020	Israel	N/A	G06N 5/02
2022/0172050	12/2021	Dalli	N/A	G06N 3/047
2022/0391663	12/2021	Zhang	N/A	G06N 3/04
2023/0199026	12/2022	Venzin	726/22	G06N 3/088
2023/0222208	12/2022	Razi	726/23	G06N 3/088
2023/0300161	12/2022	Jenks	N/A	H04L 63/1433
2023/0353587	12/2022	Bui	N/A	H04L 63/1416
2023/0362175	12/2022	Li	N/A	H04L 41/16
2023/0421578	12/2022	Mullins	N/A	H04L 63/1416
2024/0211596	12/2023	Livne	N/A	G06N 3/08
2024/0333747	12/2023	Parla	N/A	G06F 21/31
2024/0333749	12/2023	Nabeel	N/A	H04L 63/1433
2024/0403428	12/2023	Lal	N/A	G06F 21/566
2025/0097242	12/2024	Zhao	N/A	H04L 63/102

OTHER PUBLICATIONS

Kaize Ding et al.; Cross-Domain Graph Anomaly Detection; IEEE Transactions On Neural Networks and Learning Systems, vol. 33, No. 6, Jun. 2022; 2022 (Year: 2022). cited by examiner

Wenxuan He, et al.; Malicious Domain Detection via Domain Relationship and Graph Models; IEEE Xplore; 2025 (Year: 2019). cited by examiner

Shuai Zhang et al.; Attributed Heterogeneous Graph Neural Network for Malicious Domain Detection; Proceedings of the 2021 IEEE 24th International Conference on Computer Supported Cooperative Work in Design; 2021 (Year: 2021). cited by examiner

Yonghong Huang¹ et al.; Graph neural networks and cross-protocol analysis for detecting malicious IP addresses.; Springer; 2023 (Year: 2023). cited by examiner

Alexa Internet, Inc. Top sites, 2021. www.alexacom.com., Alexa Internet, Inc. Copyright 1996-2021. URL: <https://web.archive.org/web/20210811170942/https://www.alexacom.com/topsites>, (downloaded Aug. 11, 2021, by web.archive.org). cited by applicant

Chen, X., M. Chen, W. Shi, Y. Sun, and C. Zaniolo. Embedding uncertain knowledge graphs. In Proceedings of the AAAI Conference on Artificial Intelligence, vol. 33, No. 1, pp. 3363-3370, 2019. cited by applicant

CROWDSTRIKE. Falcon X: Automated Threat Intelligence, 2021. URL:

<https://web.archive.org/web/20210811202433/https://www.crowdstrike.com/endpoint-security-products/falcon-x-threat-intelligence/> (downloaded Aug. 11, 2021, by web.archive.org). cited by applicant

DOMCOP. Top 10 Million Domains, 2021. URL: <https://web.archive.org/web/20210813211408/https://www.domcop.com/top-10-million-domains>(downloaded Aug. 13, 2021, by web.archive.org). cited by applicant

Eckart, C., & Young, G. (1936). The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3), 211-218. cited by applicant

Hubbard, Dan. Cisco Umbrella: Top 1 million, Dec. 14, 2016, Updated Mar. 31, 2021, URL: <https://web.archive.org/web/20210802174758/https://umbrella.cisco.com/blog/cisco-umbrella-1-million> (downloaded Aug. 2, 2021, by web.archive.org). cited by applicant

MAJESTIC. The Majestic Million, 2021. URL: <https://web.archive.org/web/20210811005852/https://majestic.com/reports/majestic-million> (downloaded Aug. 11, 2021, by web.archive.org). cited by applicant

Perozzi, B., Al-Rfou, R. and Skiena, S. 2014. Deepwalk: Online Learning of Social Representations. In KDD. http://perozzi.net/publications/14_kdd_deepwalk.pdf. cited by applicant

PROOFPOINT. Emerging Threats Intelligence, 2021. URL: <https://www.proofpoint.com/us/products/et-intelligence> (downloaded Aug. 11, 2021, by web.archive.org). cited by applicant

Sato, R., Yamada, M., & Kashima, H. (2021). Random features strengthen graph neural networks. In Proceedings of the 2021 SIAM international conference on data mining (SDM) (pp. 333-341). Society for Industrial and Applied Mathematics. cited by applicant

Torkamani, A., B. Coskun, and B. Jeffrey. Scoring domain names at scale using k-partite graph convolution and embedding. 8th Amazon Machine Learning Conference (AMLC), 2020. cited by applicant

Tsoumakas, Grigorios, and Ioannis Katakis. “Multi-label classification: An overview.” *International Journal of Data Warehousing and Mining (IJDWM)* 3, No. 3 (2007): 1-13. cited by applicant

Wu, Felix, Amauri Souza, Tianyi Zhang, Christopher Fifty, Tao Yu, and Kilian Weinberger. “Simplifying graph convolutional networks.” In *International conference on machine learning*, pp. 6861-6871. PMLR, 2019. cited by applicant

Primary Examiner: Zand; Kambiz

Assistant Examiner: Ahmed; Mahabub S

Attorney, Agent or Firm: Atborus, PLLC

Background/Summary

BACKGROUND

(1) Domain Name System (“DNS”) is a mapping system in which Internet domain names are mapped to Internet Protocol (“IP”) addresses. Domains are one of the major attack vectors used in various cybercrime, such as spam, phishing, malware, botnets, etc. Therefore, it is essential to effectively detect and block malicious domains when combating cyber attackers.

(2) Traditional systems that attempt to detect and block malicious domains typically follow a feature based approach. While these techniques provide good performance, potential problems still exist. First, in the training phase, these traditional systems require labeled datasets large enough to ensure accuracy and coverage. However, the fickle nature of DNS makes accurate labeling an

arduous process. Second, traditional systems treat each domain individually and rely on some manually selected statistical features (e.g., number of distinct IP addresses, the standard deviation of Time to Live (“TTL”), etc.), making evasion of the detection system by attackers possible.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a transition diagram illustrating the training and use of a graph neural network (“GNN”) in which the GNN is trained to monitor for malicious domains and, upon detection of a malicious domain, provide protection to users of a cloud service, classify the malicious domain, and provide information to the users of the cloud service that indicates why the domain is determined to be malicious, in accordance with disclosed implementations.
- (2) FIG. 2 is an illustration of both known benign domains and known malicious domains, along with interconnection indications between the domains and compute instances that are monitored by the monitoring service, in accordance with disclosed implementations.
- (3) FIG. 3 is a block diagram illustrating the training of a graph neural network based on a feature matrix generated from the embeddings discussed with respect to FIG. 2, in accordance with disclosed implementations.
- (4) FIG. 4 is an example malicious domain determination process, in accordance with disclosed implementations.
- (5) FIG. 5 is an example domain monitoring process, in accordance with disclosed implementations.
- (6) FIG. 6 is an illustration of a generation of an adjacency-based similarity graph embedding, in accordance with disclosed implementations.
- (7) FIG. 7 is a graph illustrating prediction confidence scores determined between tactics and attributes for domains, in accordance with disclosed implementations.
- (8) FIG. 8 illustrates example components of a server, in accordance with disclosed implementations.

DETAILED DESCRIPTION

- (9) Disclosed are systems and methods that monitor for malicious and unauthorized behaviors, determine categories for detected malicious behaviors, determine why a domain is determined to be malicious, and provide information to users that identifies the categories and reasons as to why a domain is determined to be malicious. In some implementations, the disclosed systems and methods may be utilized to provide monitoring security to customers of a cloud service. For example, customers of a cloud service may maintain an account with the cloud service and the disclosed implementations may be utilized to protect those accounts and compute instances (“CI”) generated through those accounts, from malicious attacks and cybercrimes such as, but not limited to, spam, phishing, malware, botnets, etc.
- (10) The monitoring service, as discussed herein, may maintain a list of malicious domains, which may be determined from the monitoring service itself and/or obtained from other threat intelligence (“TI”) services (e.g., ADMINUSLabs, CRDF, Kaspersky, AlienVault, etc.) in the form of threat lists. However, threat lists received from other TI services may include false positives, which could lead to adverse operational consequences for customers of the cloud service, such as blocking of legitimate domains. Additionally, in some implementations, information about a domain received from a first TI service may conflict with information received from a second TI service. For example, a popular domain could be abused and appear malicious to some TI services. As a result, a first TI list received from a first TI service may identify the domain as benign while a second TI list from a second TI service may identify the domain as malicious. Accordingly, the disclosed implementations normalize threat lists received from TI services, improve the accuracy in

determining malicious domains, and provide information to users as to why a domain was determined to be malicious, and optionally provide an identification of a determined classification for the malicious domain. Continuing with the above example of conflicting information about a domain and/or instances in which TI services use different descriptors/categories for domains, as discussed herein, the disclosed implementations may normalize received information into a normalized schema for the domains that allows the information from each TI service to co-exist in the representations generated for the domains.

(11) As discussed further below, the disclosed implementations generate a large-scale heterogeneous graph of interactions between monitored compute instances, domains, Internet Protocol (“IP”) addresses, account identifiers (“ID”) for accounts that use a compute service to generate CIs, Canonical Name Records (“CNAME”), etc., and their diverse relationships. For example, the DNS scene may be mapped into an M-partite graph that includes multiple nodes and edges. Nodes may represent, for example, CIs, domains, IP addresses, account ID of accounts of the monitoring service, etc. Rather than rely on manually selected statistical features, as is traditionally done, in accordance with the disclosed implementations, useful information from labeled domains may be extracted by building a knowledge graph over the subcategories of domain labels. The knowledge graph will generate embeddings for all the subcategories of domains, which can be used as initial embeddings in the M-partite graph. Likewise, the knowledge graph may be further supplemented with embeddings for each of a plurality of known benign domains. Still further, message passing between different nodes of the M-partite graph helps propagate knowledge of the graph among nodes and generate meaningful embeddings for nodes corresponding to domains in which it is unknown as to whether the domain is malicious or benign.

(12) FIG. 1 is a transition diagram illustrating the training and use of a graph neural network (“GNN”), or other machine learning model, in which the GNN is trained to monitor for malicious domains and, upon detection of a malicious domain, provide protection to users of a cloud service, classify the malicious domain, and provide information to the users of the cloud service that indicates why the domain is determined to be malicious, in accordance with disclosed implementations.

(13) The monitoring service **120-1**, which may be executing on one or more computing resources **121**, may receive threat intelligence (“TI”) information **150-1** from one or more TI services **177**. As is known, there are multiple TI services **177** such as CISCO **177-1**, CRDF **177-2**, DOMCOP **177-3** through ALEXA **177-M**. Each of these TI services **177** may monitor domains to generate TI lists that include indications as to whether a domain is determined to be benign or malicious. Many of these TI services also include categories or other information in the TI lists for domains determined to be malicious. However, TI lists received from the various TI services **177** are often inconsistent and/or conflicting. Likewise, there is no consistency for designation of categories or other information included in TI lists when describing malicious domains. For example, a first TI service **177-1** may provide a single “category code” whereas a second TI service **177-2** may provide multiple categories or descriptions for a determined malicious domain that have little or no relation to the category codes used by the first TI service. Likewise, in some instances, different TI services may provide conflicting categorizations of a domain. For example, a popular domain could be abused and appear malicious to some TI services but not others. As a result, a first TI list received from a first TI service may identify the domain as benign while a second TI list from a second TI service may identify the domain as malicious.

(14) The monitoring service **120-1**, upon receiving TI lists from one or more TI services **177**, such as a first TI service **177-1**, a second TI service **177-2**, a third TI service **177-3**, through an M-th TI service **177-M**, may transform the different received TI lists into a consolidated and normalized schema **150-2**. In some examples, the various threat lists received from the different TI services **177** may be mapped into defined tactics and attributes maintained by the monitoring service, thereby normalizing the inconsistent use of information/categories utilized to describe domains that

are determined to be malicious while at the same time retaining the information received from the different TI services regarding those domains. Tactics may include, for example and not by way of limitation, initialAccess, exfiltration, reconnaissance, resourceDevelopment, impact, commandAndControl, etc. Attributes may include, for example and not by way of limitation, technique/typoSquatting, characteristics/actorControlled, intent/phishing, intent/exploitDelivery, intent/malwareDelivery, etc. Any number and/or type of tactics and/or attributes may be defined and maintained by the monitoring service. Still further, in some implementations, the monitoring service may maintain or define malware classifiers into which malware domains are classified. Malware classifiers may include, by way of example and not limitation, PittyDownloaderA, ChattyCRC, msger, Emdivi, HtDnDownLoader, POWERTON, Scanbox, etc.

(15) Other information that may be received and/or maintained by the monitoring service for domains may include ranking categories for domains (e.g., top20k/alexa, top100k/alexa, top1m/alexa, top500k/cisco, etc.) which may rank domains based on popularity/frequency of access.

(16) As part of normalization, the confidence values for each Indicator of Compromise (“IoC”) received from the various TI services **177** for each domain may be normalized to a defined scale, such as -100 (malicious) to 100 (benign). Still further, in some examples, a set of rules may be applied to assign higher or lower confidence values to certain threat types. For example, threat types relating to Crypto may be assigned a higher threat value than other threat types.

(17) Once the different content of the different TI lists received from the different TI services **177** has been consolidated and normalized into a normalized schema, the monitoring service **120-1** may generate pretrained representations for each possible normalized label type (tactics, attributes, ranking, etc.). As discussed herein, the pretrained representations are additive, therefore by adding two or more of them together, all information relating to the representations is retained and may be utilized as labels for the representation.

(18) A graph that includes nodes and edges of the DNS scene may also be generated. As discussed further below, in some implementations, nodes may be defined by CI, domain, IP address, account identifier of accounts of the monitoring service, etc. Edges may be defined as the interconnections between domains and CI that are monitored by the monitoring service, interconnections between the domain and IP address, interconnections between the domain and the corresponding second level domain and top-level domain (“TLD”) pairs, interconnections between the domain and CNAME record, interconnections between the domain and account Id of the account that maintains the CI, etc. As discussed further below, a variety of techniques may be used to define the graph, the positioning of the nodes, and the corresponding edges.

(19) The pretrained representations of each possible label may be utilized to generate initial representations (e.g., distributed vector representations), also referred to herein as initial embeddings, for each node of the graph (both benign domains and malicious domains), as in **150-3**. As discussed further below, there are a variety of techniques that may be utilized to generate the initial embeddings for nodes of the graph.

(20) The graph of known nodes and edges may then be used to train a GNN and classifier to predict whether a node is benign or malicious, and determine the tactics and attributes of the node if the node is determined to be malicious, as in **150-4**. For example, as discussed further below, message passing techniques, which update the state of each node based on information received from neighbor nodes, may be used to tune and scale the GNN for use on a large graph, such as a graph of the DNS scene that has billions of nodes and trillions of edges. In some implementations, the GNN may have two layers such that message passing goes two steps into the graph. In other implementations, the GNN may have fewer or additional layers and there are fewer or additional levels of message passing.

(21) The last stage of the GNN, after message passing, may be a classifier, such as a multi-labeled classifier, that is trained to process the final embeddings generated by the GNN to determine if the

domain represented by the final embedding is malicious or benign and/or to classify the domain if the domain is determined to be malicious (e.g., assign the domain to a defined classification, assign tactics, attributes, etc.). The classifier may be any of a variety of classifiers. For example, the classifier may be a Binary Relevance classifier, Neural Network classifier, a Random Forest classifier, etc.

(22) By using pretrained representations of labels that are included in the embeddings of each node that is known to correspond to a benign or malicious domain, further training of the middle layers of the GNN is not required, thereby reducing the computing time and complexity needed to train the GNN. For example, the pretrained representations may be propagated as nodes' seed features via a message passing process between layers of the GNN. Message passing results in concise context representation in the neighborhood of a domain in the graph that helps explain the rationale behind each prediction as well as enabling the monitoring service to predict specific tactics and attributes associated with each domain name or IP address. This structure also increases the accuracy of detection performance in determining if an unknown domain is malicious or benign.

(23) In some implementations, the GNN may be configured to process embeddings of the graph periodically (e.g., hourly, daily, weekly) and update the determinations for domains represented in the graph. As will be appreciated, the entire DNS scene may be processed as domains may change from benign to malicious and/or from malicious to benign and new domains may be added to the DNS scene. In other implementations, only portions of the DNS scene may be periodically processed, only newly identified domains may be processed, such as in real time or near real time, segments of the DNS scene may be processed on a rolling basis, etc.

(24) Regardless of frequency, the disclosed implementations may be used to monitor **150-5** and determine the state (malicious or benign) of existing domains and/or newly identified or unknown domains **111** accessed by compute instances hosted by the cloud service **120-2** for different users **101** of the cloud service. If a domain **111** that is unknown to the monitoring service **120-1** is accessed by a compute instance of a user **101**, the monitoring service may process the unknown domain **111** with the GNN to determine whether the domain is benign or malicious, as in **150-6**. For example, the unknown domain may be represented as a node in the graph, the graph, or a portion thereof, processed by the GNN to generate an embedding for the unknown domain (e.g., through message passing/propagation). The generated embedding for the unknown domain may then be analyzed to determine if the domain is benign or malicious. The GNN may also identify the tactics, attributes, classification, etc. of domains determined to be malicious.

(25) If the domain is malicious, the monitoring service **120-1**, at **150-7**, may take action on behalf of the user **101** and block access with the domain. Likewise, because the GNN is able to provide context about the domain (e.g., tactics, attributes, classification), at **150-8** the monitoring service **120-1** may provide such information to the user **101** so the user has insight as to why the domain request was blocked and the domain identified as malicious.

(26) The system may also include computing resource(s) **121**. The computing resource(s) **121** may be remote from users **101** and/or TI services **177**. Likewise, the computing resource(s) **121** may be configured to communicate over a network **102** with the TI services **177**, users **101** and/or domains **111**.

(27) As illustrated, the computing resource(s) **121** may be implemented as one or more servers **121(1)**, **121(2)**, . . . , **121(N)** and may, in some instances, form a portion of a network-accessible computing platform implemented as a computing infrastructure of processors, storage, software, data access, and so forth that is maintained and accessible by components/devices of the system via a network **102**, such as an intranet (e.g., local area network), the Internet, etc. The computing resources **121** may host one or more of the monitoring service **120-1** and/or the cloud service **120-2**, execute a GNN, etc., as discussed herein.

(28) The server system(s) **121** does not require end-user knowledge of the physical location and configuration of the system that delivers the services. Common expressions associated for these

remote computing resource(s) **121** include “on-demand computing,” “software as a service (SaaS),” “platform computing,” “network-accessible platform,” “cloud services,” “data centers,” and so forth. Each of the servers **121(1)-(N)** include a processor **117** and memory **119**, which may store or otherwise have access to a monitoring service **120-1** and/or the cloud service **120-2**, as described herein. An example server is discussed further below with respect to FIG. 8.

(29) The network **102**, and each of the other networks discussed herein, may utilize wired technologies (e.g., wires, USB, fiber optic cable, etc.), wireless technologies (e.g., radio frequency, infrared, NFC, cellular, satellite, Bluetooth, etc.), or other connection technologies. The network **102** is representative of any type of communication network, including data and/or voice network, and may be implemented using wired infrastructure (e.g., cable, CAT6, fiberoptic cable, etc.), a wireless infrastructure (e.g., RF, cellular, microwave, satellite, Bluetooth, etc.), and/or other connection technologies.

(30) FIG. 2 is an illustration of both known benign domains and known malicious domains, along with interconnection indications between the domains and compute instances that are monitored by the monitoring service, in accordance with disclosed implementations.

(31) As illustrated in FIG. 2, domains **211**, both known benign domains **211-B1**, **211-B2**, **211-B3**, through **211-BN** and known malicious domains **211-M1**, **211-M2**, **211-M3**, through **211-MR**, along with tactics, attributes, rankings, classifications, etc., associated with those domains, may be used as part of generating training data for training a GNN, or other machine learning model. For example, some very highly ranked domains, such as amazon.com **211-B1**, google.com **211-B2**, facebook.com **211-B3** through other.com **201-BN** may be well known and highly accessed domains that, from monitoring, ranking, common knowledge, etc., are known to be benign domains **211** (i.e., not malicious). Such well known domains **211** may have a disposition **231** of benign associated with the domain that is maintained by the monitoring service. In some implementations, rankings assigned to those domains **211** may also be maintained and included in the representations of those domains.

(32) Likewise, there are other domains, such as malice.com **211-M1**, attacker.com **211-M2**, attack.com **211-M3**, through bad.com **211-MR** that are well known and established to be malicious domains **211**. For each of the known malicious domains **211-M1** through **211-MR** the monitoring service may maintain a disposition **231** of threat (or other indicator that the domain is malicious), along with determined tactics **241** and/or attributes **251** for each domain. As discussed above, tactics and attributes may be normalized based on categories or other classifiers assigned to the domains by other TI services and/or by the management service.

(33) In the illustrated example, the malicious domain **211-M1** of malice.com has a disposition **231-1** of Threat, a tactics **241-1** of initialAccess and an attribute of **251-1** of intent/payment. The malicious domain **211-M2** of attacker.com has a disposition **231-2** of Threat, a tactics **241-2** of initialAccess and an attribute of **251-2** of intent/payment. The malicious domain **211-M3** of attack.com has a disposition **231-3** of Threat, a tactics **241-3** of initialAccess and an attribute of **251-3** of intent/malwareDel. The malicious domain **211-MR** of bad.com has a disposition **231-R** of Threat, a tactics **241-R** of initialAccess and an attribute of **251-R** of intent/malwareDel.

(34) As discussed herein, because the graph includes both benign domains and malicious domains, which are therefore incorporated and used with a GNN, the traditional need for manually assigned weights to combine lists of benign domains and malicious domains or requiring assumptions about the validity of lists (or the corresponding domains) is eliminated. This benefit provides a technical improvement over existing systems and eliminates the need for manually designed and handcrafted features.

(35) In addition to domains **211** and information about those domains, such as tactics **241** and/or attributes **251**, edges **261** may be established that indicate access requests between nodes and CIs such as CI-1 **211-C1**, CI-2 **211-C2**, CI-3 **211-C3**, through CI-P **211-CP**.

(36) In accordance with the disclosed implementations, data extracted from DNS records regarding

known benign domains and known malicious domains, along with information known about CIs and accounts for those CIs, and interconnections therebetween, may be used to generate a knowledge graph. For example, in some implementations, a knowledge graph **200** may be established in which nodes are defined as: CI: identifier for a compute instance that was observed querying a domain, full domain: Tuple of (subdomain, second-level domain, top-level domain (“TLD”)), second-level domain and TLD pair: Tuple of (second-level domain, TLD), IP address: IPv4, IPv6 address to which a domain resolves, CNAME, and account identifier (“ID”): the account ID of the cloud service that owns or generated the CI, and accounts.

(37) Likewise, edges **261** between nodes **211** of the graph may be defined as: (full domain, CI): There is an edge between the CI ID and the full domain queried by that instance, (Full domain, IP): There is an edge between the full domain and IP if the full domain is resolved to that IP address, (Full domain, Second-Level Domain, and TLD): There is an edge between every full domain and its (second-level domain, TLD) pair, (Full domain, CNAME): There is an edge between the full domain and the CNAME record, and (Full domain, account): There is an edge between the full domain and the account ID.

(38) As an example illustration, the DNS scene may be formulated as an attributed graph denoted by $G=(V, E)$, where $V=\{v_{\text{sub.1}}, \dots, v_{\text{sub.N}}\}$ is the set of N nodes in graph G , and E is the set of edges. More specifically, by extracting the information from the DNS data, nodes and edges of the graph may be defined and utilized to generate an M -partite graph. In the discussed example, $M=6$. In other implementations the number of nodes, and thus the corresponding number M for the M -partite graph may be higher or lower. $A \in \mathbb{R}^{\text{sup.N} \times \text{sup.N}}$ may be denoted as the corresponding adjacency matrix whose element $A_{\text{sub.UV}} \in \{0,1\}$ indicates the existence of edge $e_{\text{sub.UV}}$ that connects nodes U and V . $X=\{x_{\text{sub.1}}, \dots, x_{\text{sub.N}}\}$ with $x_{\text{sub.i}} \in \mathbb{R}^{\text{sup.d}}$ may also be denoted as the d -dimensional node feature for node $v \in V$. Each node may also be associated with a set of C labels $\{y_{\text{sub.1}}, y_{\text{sub.2}}, \dots, y_{\text{sub.C}}\}$ with $y_{\text{sub.i}} \in \{0,1\}$.

(39) As will be appreciated, not all nodes established from a DNS scene are known and therefore may not have corresponding labels, such as those discussed above with respect to the malicious domains. Accordingly, labeled nodes, such as the known benign domains and the known malicious domains, and the data corresponding thereto, may be denoted as $V_{\text{sub.L}}$, where the corresponding labels are denoted as $Y_{\text{sub.L}}=\{y_{\text{sub.V}}\}_{\text{sub.V} \in V_{\text{sub.L}}}$. Comparatively, the set of unlabeled nodes may be defined as $V_{\text{sub.U}}:=V \setminus V_{\text{sub.L}}$. In the illustrated example, semi-supervised multi-label node classification may then be utilized. For example, given A , X and the label information of the $V_{\text{sub.L}}$, the goal may be to infer the label of nodes in $V_{\text{sub.U}}$ by learning a classification function f with parameter θ . The overall problem may then be formulated as:

$$(40) \quad \min_{\theta \in \mathbb{R}^d} \hat{f}(\theta) = L(f(A, X; \theta), Y_L) = \sum_{v_i \in V_L} L(f(A, x_i; \theta), \{y_i\}) \quad (1)$$

where $f(A, x_{\text{sub.i}}; \theta)$ and $y_{\text{sub.i}}$ are the predicted and the true labels of node $v_{\text{sub.i}}$, and $L(\cdot, \cdot)$ stands for a loss function (e.g., Multi-label soft margin function loss). A GNN may then be trained to solve the problem. While the below discussion focuses on using a GNN to solve the problem, in other implementations, other forms of machine learning models may be utilized.

(41) GNNs learn compact, low-dimensional representations (embeddings) for nodes such that representation captures the nodes local network neighbor as well as the nodes features. Formally, the k -th layer of a GNN obeys the following propagation rule to aggregate the neighboring features:

$$h_{\text{sub.i}}^{\text{sup.(k)}} = g^{\text{sup.(k)}}(\{W^{\text{sup.(k-1)}} h_{\text{sub.j}}^{\text{sup.(k-1)}} \tilde{A}_{\text{sub.ij}}, \forall j \in N_{\text{sub.i}}\}, \forall i \in [N])$$

where $h_{\text{sub.i}}^{\text{sup.(k)}}$ denotes the feature vector of node i at layer k , $h_{\text{sub.i}}^{\text{sup.(0)}}=x_{\text{sub.i}}$ is the input feature for node i , $g^{\text{sup.(k)}}$ denotes a composite mapping function (e.g., an activation function), and $W^{\text{sup.(k-1)}}$ is a trainable weight matrix at $(k-1)$ -th layer. $\tilde{A}$ denotes a linear mapping of the adjacency matrix A with some sparsity pattern, and $\tilde{A}_{\text{sub.ij}}$ is the (i,j) -th entry of

$\tilde{A}$. N.sub.i indicates immediate neighbors of node i, including the node itself ($i \in N$.sub.i).

(42) An example of a GNN architecture is a graph convolutional network (“GCN”), which considers learning on graph structures using convolutional operations, and works well with embeddings. In GCN, the generic propagation rule is specified as:

$$(43) h_i^{(k)} = \text{ReLU} \left(\sum_{j \in N_i} W^{(k-1)} h_j^{(k-1)} \tilde{A}_{ij} \right)$$

where $\text{ReLU}(\cdot)$ is the ReLU activation function. $\tilde{A}$.sub.ij is the i-th row of $\tilde{A}$. In the illustrated example, $\tilde{A}$ is the normalized adjacency matrix $\tilde{A} = \{\text{circumflex over (D)}\}^{\text{sup.}-1/2} \hat{A} \{\text{circumflex over (D)}\}^{\text{sup.}-1/2}$ with $\hat{A} = A + I$, and $\{\text{circumflex over (D)}\}^{\text{sup.}-1/2}$ is the inverse square root of the degree matrix $\{\text{circumflex over (D)}\}$.sub.ij=0 if $i=j$ $\{\text{circumflex over (D)}\}$.sub.ij=1.^{sup.TA.sub.:},i. Based on the above, the standard form of a GCN can be represented as:

$$H^{\text{sup.}(k)} = \text{ReLU}(\tilde{A} H^{\text{sup.}(k-1)} (W^{\text{sup.}(k-1)})^{\text{sup.T}})$$

where

$$(44) H^{(k)} = [(h_1^{(k)})^T; \dots; (h_N^{(k)})^T].$$

(45) After obtaining the final feature representation from the output of the K-th layer, the forward model $f(A, X; \theta)$ with $\theta = \{W\}$ in equation (1) above, the problem can be represented as:

$$f(A, X; \theta) = \text{SoftMax}(\tilde{A} H^{\text{sup.}(K)} W^{\text{sup.}(K)}),$$

Wherein $\text{softmax}(\cdot)$ is the SoftMax function.

(46) Because typical GNNs cannot scale to extremely large graphs, such as a DNS graph with billions of nodes and trillions of edges, instead of learning a (K-1)-layers model, the above discussed convolutional methods are simplified into semi-supervised message passing procedures, and then the classification model (i.e., K-layer) is learned based on the output of the message passing. The simplified graph neural network may include two stages: feature propagation in each relation and final model predictions (multi-label classification in our case). Feature propagation aggregates messages received from neighboring nodes in each relation and uses this information to iteratively update the nodes own representation. For example, given a node i, its vector representation h .sub.i.^{sup.(k)} at layer k, and its neighborhood N .sub.m(i) over m-th relation, a message passing update can be written as:

$$(47) h_i^{(k)} = \frac{1}{M} h_i^{(k-1)} + \sum_{m=1}^{M-1} \frac{1}{\sum_{j \in N_m(i)} \{\text{circumflex over (D)}\}^{\text{sup.}-1/2}} \sum_{j \in N_m(i)} h_j^{(k-1)}.$$

(48) As illustrated, an embedding may be generated for each labeled node. For example, the benign domains, represented as nodes **211-B1**, **211-B2**, **211-B3**, through **211-BN** may be represented as embeddings **270-B1**, **270-B2**, **270-B3**, through **270-BN**, respectively. CIs, represented as nodes **211-C1**, **211-C2**, **211-C3**, through **211-CP**, may be represented as embeddings **270-C1**, **270-C2**, **270-C3**, through **270-CP**. The benign domains, represented as nodes **211-M1**, **211-M2**, **211-M3**, through **211-MR** may be represented as embeddings **270-M1**, **270-M2**, **270-M3**, through **270-MR**, respectively. As noted above, the embeddings may be formed from predefined representations for each possible label that may be used in accordance with the disclosed implementations.

(49) Referring to FIG. 3, the final embeddings of the graph may be represented as a feature matrix **301** that is used to train a multi-label classifier **304**, such as a Binary Relevance classifier, a Neural Network classifier, a Random Forest classifier, etc., which may have multiple hidden layers **302-1**, **302-2**, etc., to classify nodes of the graph, such as newly identified nodes into one or more categories, such as malicious **304-1**, phishing **304-2**, sinkholed **304-3** through malware host **304-X**. Any number of classifications may be defined and used with the multi-labeled classifier to classify a domain and those provided with respect to FIG. 3 are illustrated as examples only. As discussed herein, the classifier and the information determined therefrom may be used to determine not only whether a domain is malicious or benign but also tactics, attributes, categories, etc., for a malicious domain. Such information for a malicious domain may be provided to operators to determine what actions to take with respect to the domain, may be used to cause one or more automated actions to be performed with respect to the domain (e.g., deny access with the domain), and/or to provide

additional information to a user as a basis as to why a domain is determined malicious.

(50) FIG. 4 is an example malicious domain determination process **400**, in accordance with disclosed implementations. The example process **400** may be performed periodically (e.g., hourly, daily, weekly) and may be performed on, for example, the entire DNS scene, segments of the DNS scene, select domains, etc. In some implementations, the example process may be performed daily on the entire DNS scene and then performed in real-time or near real-time for any newly identified domains.

(51) The example process **400** begins by generating a normalized schema for domains that are to be processed by the example process **400**, as in **402**. As discussed above, in some examples, TI lists may be received from one or more TI services. In other implementations, the example process may maintain its own TI list. Regardless of the source, the normalized scheme of domains may include some domains of the DNS scene or all domains of the DNS scene. As discussed above, for domains that are known to be malicious, the normalized schema may also include labels for tactics, attributes, and/or categories for each of those malicious domains to provide context as to why the domain is determined malicious. Likewise, the tactics, attributes, and/or categories may be selected from a defined list of tactics, attributes, and/or categories. In addition, defined representations may be generated for each of the defined labels (tactics, attributes, categories), as in **404**. Providing defined representations for the list of labels allow for embeddings to be updated through message passing, as discussed herein, without losing context or information about the domains.

(52) The example process **400** may then generate a graph that includes nodes and edges, as in **406**. As discussed above, the nodes of the graph may represent, for example, CIs, domains, IP addresses, account ID of accounts of the monitoring service, etc. Likewise, edges of the graph may represent, for example, interconnections between domains and CIs that are monitored by the monitoring service, interconnections between the domain and IP address, interconnections between the domain and the corresponding second level domain and TLD pairs, interconnections between the domain and CNAME record, interconnections between the domain and account ID of the account that maintains the CI, etc.

(53) Embeddings for the domains of the graph may also be generated, as in **408**. As discussed above, embeddings may be generated for domains that are known to be benign or known to be malicious based on the information known about those domains, which may include the representations for the labels associated with those domains. In comparison, for domains that are unknown, the embedding may only include the limited information known about the domain (e.g., domain name, IP address, etc.)

(54) The embeddings of the graph may then be processed by a GNN, through the use of message passing, to update the embeddings for each node/domain of the graph based on information learned about the neighboring nodes of the graph. Message passing may be determined based on the number of layers of the GNN. For example, in some implementations, messages may be passed two layers. In other implementations, message passing may be longer or shorter. When message passing is complete, each node/domain of the graph has a final embedding that has been generated by updating the initial embedding generated for the node based on messages received from neighboring nodes.

(55) The example process **400** may then process the final embeddings of the domains with a classifier, such as a multi-layer classifier, to determine whether the domain is benign or malicious, as in **412**. Finally, as discussed herein, for domains determined to be malicious, the final embedding generated for those domains may be further processed to determine the tactics, attributes, and/or classification of the domain, as in **414**. As discussed herein, because tactics, attributes, and/or classifications are labels determined from a defined set of tactics, attributes, and/or classifications, each of which have a defined representation, the labels generated through message passing retain context and can be used to determine a tactic, context, classification for a domain/node based on the final embedding generated for that domain/node.

(56) FIG. 5 is an example domain monitoring process **500**, in accordance with disclosed implementations. The domain monitoring process may be periodically or continuously performed and may be used to monitor any number of CIs, as discussed herein.

(57) The example process **500** begins by monitoring one or more CI(s), as in **502**. As the example process is monitoring the one or more CI(s), a determination may be made as to whether an access between a monitored CI and a domain is attempted, as in **504**. If it is determined that an access is not attempted, the example process **500** returns to block **502** and continues. If it is determined that an access is attempted, it is determined whether the domain related to the attempted access has been determined benign, as in **506**. For example, and as discussed throughout, the disclosed implementations may periodically (e.g., hourly, daily, etc.) process domains to determine whether those domains are benign or malicious.

(58) If it is determined that the domain for which the access is attempted is determined to be benign, the access is allowed, as in **508**.

(59) If it is determined that the domain is not known to be benign, it is determined whether the domain is known to be malicious, as in **510**. If it is determined that the domain is not known to be malicious (i.e., the domain is unknown), the malicious domain determination process **400** (discussed above) is performed, as in **512**, and the example process **500** returns to block **504** and continues. However, if it is determined that the domain is known to be malicious, the access attempt is blocked, as in **514**, and a notification that includes a basis as to why the domain is determined to be malicious is generated, as in **516**. As discussed above, the basis as to why a domain is determined malicious may be determined from labels included in the final embedding generated for the domain and provided as the basis as to why the domain is malicious.

(60) As is known, embedding features help GNNs capture node distance information regarding their relative positions in a graph. For example, in the DNS graph, considering two compute instances, if the two instances have never queried a known malicious domain, it is more likely that the other unknown domains the two instances have queried are also non-malicious and their embedding features should be close to each other in the embedding graph. In comparison, if one CI has mostly queried known malicious domains and the other CI has mostly queried known malicious domains, the node features representative of those two nodes should be far from each other in the node graph.

(61) However, because of the size of the DNS graph, which includes billions of nodes and trillions of edges, typical feature initialization methods do not perform well. For example, the one-hot method, due to the size of the DNS graph, will initialize feature vectors with extremely high dimensions, which does not scale to extremely large graphs with message passing. Similarly, the eigen and deepwalk initialization methods, which generate features by matrix decompositions, will require tremendous computation resources to perform matrix decomposition for a graph with as many nodes and edges as a DNS graph considered in the disclosed implementations.

(62) Accordingly, the disclosed implementations may use one of the following techniques to initialize node features.

(63) In a first example, referred to herein as label co-occurrence Graph, a graph is defined as $G=(L,E)$, where $L=\{L.sub.1, L.sub.2, \dots, L.sub.C\}$ represents the set of N nodes and E represents the set of connections between any two nodes (i.e., edges), and the adjacency matrix $\tilde{A}$ is a conditional probability matrix by setting $A.sub.ij=P(L.sub.i|L.sub.j)$, wherein P is calculated through the training set. Because $P(v.sub.i|v.sub.j)=P(v.sub.j|v.sub.i)$, namely $A.sub.ij=A.sub.ji$, in some examples A can be symmetrized by

(64) $\tilde{A}' = \frac{1}{2}(A + A^T)$.

(65) For example, FIG. 6 illustrates a representation of a symmetrical adjacency matrix **601**, in accordance with the disclosed implementations. As illustrated, through the adjacency matrix, the representations illustrate, for embeddings, the relationships between threat **602** identifiers and benign **603** identifiers, attributes **601-1**, ranks **601-2**, and tactics **601-3**.

(66) In this example, embedding vectors for labels include (a) preserving the neighborhood proximity between connected node pairs, (b) keeping the similar albeit non-adjacent nodes relatively close to each other, (c) ensuring that dissimilar and non-adjacent nodes are placed far away from each other, (d) preserving the high-order structural relationship in the embedding space, and (e) representing the graph topology.

(67) Following the above guidelines, three different approaches may be utilized to generate embedding vectors. The first is directly performing matrix decomposition over the adjacency matrix, so that the complex label information can be embedded into a low dimensional representation. The second is adjacency-based similarity embedding, which captures the rich semantic relations, the spatial or temporal dependencies among labels. The third is to transform the label correlations into a knowledge graph with different relations, and then learn the embeddings via a link prediction task.

(68) To achieve fast network embedding, a randomized truncated Singular Value Decomposition (“randomized tSVD”) may be utilized, which offers significant improvements of tSVD with a strong approximation guarantee. To enable randomized tSVD, Q and d orthonormal columns are determined—i.e., $U \approx QQ^T U$. Assuming Q has been found, H can be defined as $H = Q^T U$, which is a small matrix ($d \times N$) which can be efficiently decomposed. Thus, $H = S^T \Sigma V^T$ for S^T , V^T orthogonal and Σ diagonal. Finally, U can be decomposed as $U \approx QQ^T U = (QS^T \Sigma V^T)$ and the final output embedding matrix is:

$$R = QS^T \Sigma V^T$$

and each row of R represents one node's embedding e .

(69) For example, and again referring to FIG. 6, through sparse matrix factorization graph embedding, the symmetrical graph can be factorized into an embedding matrix **610** in which each row represents the embedding of a node of the graph. Finally, using for example, cosine similarity, the embedding matrix **610** may be reproduced as a similarity matrix **620** that illustrates the correlations between the threats, attributes, rankings, tactics, etc.

(70) As an alternative to Sparse Matrix Factorization Graph Embedding, or in addition thereto, Adjacency-Based Similarity Graph Embedding may be utilized in which each node (label) in the graph is represented as word embeddings of the label. For categories whose names include multiple words, the label representation o_i may be obtained as the average of embeddings for all words for the category.

(71) A neural network may then be applied to map the pretrained embedding of each label to a semantic embedding space and produce the label embedding $e_i = \Phi(o_i)$, where Φ denotes the neural network which may consist of three fully-connected layers followed by Batch Normalization (“BN”) and ReLU activation. The goal in such a configuration is to achieve the optimal label embedding set such that $\cos(e_i, e_j)$ is close to $\tilde{A}_{ij}$ for all i, j , where $\cos(e_i, e_j)$ denotes the cosine similarity between e_i and e_j . Thereby, the objective function is defined as:

$$(72) L_{ge} = \frac{1}{N} \sum_{i=1}^N \frac{1}{N} \sum_{j=1}^N \left(\frac{e_i^T e_j}{\|e_i\| \|e_j\|} - \tilde{A}_{ij} \right)^2$$

where L_{ge} denotes the loss of graph embedding.

(73) As another alternative, Knowledge Graph Embedding may also be considered to determine the correlation between malicious categories. The definition of an Uncertain Knowledge Graph can be specified such that an Uncertain Knowledge Graph represents knowledge as a set of relations (R) defined over a set of entities (E). The Uncertain Knowledge Graph may consist of a set of weighted triples $G = \{(l, s, l)\}$. For each pair (l, s, l) , where $l = (h, r, t)$ is a triple representing a relations fact where $h, t \in E$ (the set of entities) and $r \in R$ (the set of relations), and $s \in [0, 1]$ represents the confidence score for this relation fact to be true.

(74) Example relations may include, but are not limited to, “Full domain—CI—Full domain,” “Full

domain—IP—Full domain,” “Full domain—account—Full domain,” etc. “Full domain—CI—Full domain” indicates that all domains are queried by the same CI. “Full domain—IP—Full domain” indicates that all domains are resolved to the same IP address. “Full domain—account—Full domain” indicates that all domains are queried by the same account. As will be appreciated, any of a variety of relations may be determined and those provided are for example purposes only.

(75) The confidence score $s_{sub.l} \in [0,1]$ can be interpreted as the probability of existence of a link. For example: 1. (malwareFamilies:DownExecute, IP, disposition:threat): 1 2.

(malwareFamilies:ChattyCRC; IP, tactics:commandAndControl): 0.49 3. (tactics:initialAccess, account, attributes:intent/malwareDelivery): 0.213 4. (tactics:exfiltration, CI, tractics:commandAndControl): 0.21

(76) In the above first example, it means that among the domain connected via IP addresses, if a domain with the label “malwareFamilies:DownExecute,” the probability of the domain being malicious (i.e., with the label “disposition:threat”) is 1. Thus, based on the definition of uncertain knowledge graph, the domain labels can be formalized as a link prediction task in which high-confidence triples of the form (crowdStrike, label, commandAndControl) can be predicted using the learned embeddings.

(77) Given an Uncertain Knowledge Graph G, the embedding model attempts to take advantage of patterns in the knowledge graph whereby certain high-supported relations may equivalently be represented by short indirect paths in the graph. Over a large knowledge graph, these patterns become encoded in the embeddings for entities and relations, enabling the interface of new high-confidence triples.

(78) Uncertain Knowledge Graphs need to explicitly model the confidence score for each triple and compare the prediction with the true score. Accordingly, the plausibility of triples may be defined and modeled, which can be considered as an unnormalized confidence score. Given a triple $l=(h,r,t)$ and their embeddings (h,r,t) , the plausibility parameter of (h,r,t) can be defined as $g(l)=r.Math.(h \circ t)$ where $\circ$ is the element-wise product, and $.Math.$ is the inner product. This function captures the relatedness between embeddings h and t under the condition of relation r . The resulting score may be mapped to the interval $[0, 1]$ through a logistic function:

$$(79) \quad (x) = \frac{1}{1 + e^{-(wx + b)}}$$

where w is a weight and b is a bias. The final predicted confidence score is thus:

$$f(l)=\Phi(g(l))$$

(80) For every triple used in training we may also consider negative samples/triple by corrupting the tail node and re-sampling a random node with the confidence score of 0. Thus, the total loss function is the sum of squared errors between the prediction $f(l)$, and confidence score $s_{sub.l}$ for each triple:

$$(81) J = \sum_{l \in L^+} .Math. .Math. f(l) - s_l .Math. ^2 + \sum_{l \in L^-} .Math. .Math. f(l) .Math. ^2$$

where $L_{sup.+}$ is the set of observed relation facts and $L_{sup.-}$ is the sampled set of negative triples (i.e., unseen relations). The domain labels may then be formalized as a link prediction task in which high-confidence triples are predicted using the learned embeddings.

(82) FIG. 7 is a graph 700 illustrating the final predicted confidence scores $f(l)$ for a domain based on the tactics and attributes determined for the domain, in accordance with disclosed implementations.

(83) In the illustrated graph 700, which indicates the tactics of initialAccess 702 through impact 704, and the attributes of technique/dga 706, characteristic/compromised 708, through intent/malwareDel 710, that the disclosed implementations determined the probabilities that Relation-IP 703-1 has a final predicted probability $f(l)$ of 0.7 that the node is malicious, Relation-account 703-2 has a final predicted probability $f(l)$ of 0.8 that the node is malicious, Relation-CI2 703-3 has a final predicted probability $f(l)$ of 0.73 that the node is malicious, and Relation-CI2 703-4 has a final predicted probability $f(l)$ of 1 that the node is malicious. In the illustrated

example, the scores of these triples provide the degree of association or correlation between two domains labels in terms of different relations between them. The score of these triples may not be for a domain but rather for domain labels.

(84) As discussed above, regardless of the technique used to define initial embeddings and/or the GNN, through the disclosed implementations, a monitoring service may monitor and protect CI from malicious domains and, when malicious domains are identified and blocked for a CI, the user or account associated with that CI may be provided information as to why the domain was blocked. For example, the monitoring service may provide a notification to the account/user associated with the CI notifying the user that the CI attempted to access a malicious domain and that the access was blocked. In addition, the notification may indicate a malicious classification for the domain (e.g., phishing, sinkholed, etc.), tactics determined for the domain, attributes determined for the domain, etc.

(85) FIG. 8 is a block diagram conceptually illustrating example components of a remote computing device, such as a remote server **820** that may be used to train and/or use a GNN, upon which the monitoring service is executing, and/or upon which the compute service is executing. Multiple such servers **820** may be included in the system.

(86) Each of these server(s) **820** may include one or more controllers/processors **814**, that may each include a central processing unit (CPU) for processing data and computer-readable instructions, and a memory **816** for storing data and instructions. The memory **816** may individually include volatile random access memory (RAM), non-volatile read only memory (ROM), non-volatile magnetoresistive (MRAM) and/or other types of memory. Each server may also include a data storage component **818**, for storing data, controller/processor-executable instructions, training data, labels, graphs, etc. Each data storage component may individually include one or more non-volatile storage types such as magnetic storage, optical storage, solid-state storage, etc. Each server may also be connected to removable or external non-volatile memory and/or storage (such as a removable memory card, memory key drive, networked storage, etc.), internal, and/or external networks **850** (e.g., the Internet) through respective input/output device interfaces **832**.

(87) Computer instructions for operating each server **820** and its various components may be executed by the respective server's controller(s)/processor(s) **814**, using the memory **816** as temporary "working" storage at runtime. A server's computer instructions may be stored in a non-transitory manner in non-volatile memory **816**, storage **818**, and/or an external device(s). Alternatively, some or all of the executable instructions may be embedded in hardware or firmware on the respective device in addition to or instead of software.

(88) Each server **820** includes input/output device interfaces **832**. A variety of components may be connected through the input/output device interfaces. Additionally, each server **820** may include an address/data bus **824** for conveying data among components of the respective server. Each component within a server **820** may also be directly connected to other components in addition to (or instead of) being connected to other components across the bus **824**. Each server may also include one or more trained GNNs **836**, the monitoring service **820-1**, and/or the cloud service **820-2**, as discussed herein. In some implementations, the GNNs **836** may be included in and part of the monitoring service **820-1** or may be separate from but accessible by the monitoring service **820-1**. In other examples, the monitoring service **820-1** and/or the GNNs **836** may be included in and part of the cloud service **820-2**. In still other examples, the cloud service **820-2** and the monitoring service **820-1** may be independent and execute on different computing resources. In such implementations, the monitoring service **820-1** may access or otherwise monitor traffic and/or instance of CIs generated by the cloud service **820-2**.

(89) The components of the server(s) **820**, as illustrated in FIG. 8, are exemplary, and may be located as a stand-alone device or may be included, in whole or in part, as a component of a larger device or system.

(90) The above aspects of the present disclosure are meant to be illustrative. They were chosen to

explain the principles and application of the disclosure and are not intended to be exhaustive or to limit the disclosure. Many modifications and variations of the disclosed aspects may be apparent to those of skill in the art. Persons having ordinary skill in the field of computers, communications, and machine learning should recognize that components and process steps described herein may be interchangeable with other components or steps, or combinations of components or steps, and still achieve the benefits and advantages of the present disclosure. Moreover, it should be apparent to one skilled in the art that the disclosure may be practiced without some or all of the specific details and steps disclosed herein.

(91) Aspects of the disclosed system may be implemented as a computer method or as an article of manufacture such as a memory device or non-transitory computer readable storage medium. The computer readable storage medium may be readable by a computer and may comprise instructions for causing a computer or other device to perform processes described in the present disclosure. The computer readable storage media may be implemented by a volatile computer memory, non-volatile computer memory, hard drive, solid-state memory, flash drive, removable disk, and/or other media. In addition, components of one or more of the modules and engines may be implemented in firmware or hardware.

(92) Disjunctive language such as the phrase “at least one of X, Y, or Z,” or “at least one of X, Y and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., may be any of X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain implementations require at least one of X, at least one of Y, or at least one of Z to each be present.

(93) Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” or “a device operable to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

(94) Language of degree used herein, such as the terms “about,” “approximately,” “generally,” “nearly” or “substantially” as used herein, represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “about,” “approximately,” “generally,” “nearly” or “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount.

(95) Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey in a permissive manner that certain implementations could include, or have the potential to include, but do not mandate or require, certain features, elements and/or steps. In a similar manner, terms such as “include,” “including” and “includes” are generally intended to mean “including, but not limited to.” Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more implementations or that one or more implementations necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular implementation.

(96) Although the invention has been described and illustrated with respect to illustrative implementations thereof, the foregoing and various other additions and omissions may be made therein and thereto without departing from the spirit and scope of the present disclosure.

Claims

1. A computer-implemented method, comprising: receiving, from a first threat information service, a first threat list that indicates a first plurality of domains that are determined to be malicious; receiving, from a second threat information service, a second threat list that indicates a second plurality of domains that are determined to be malicious; wherein: the first threat list includes a first set of categories to categorize domains of the first plurality of domains; the second threat list includes a second set of categories to categorize domains of the first plurality of domains; and the first set of categories is different than the second set of categories; combining the first threat list and the second threat list into a normalized schema that includes the first plurality of domains and the second plurality of domains; including, in the normalized schema, a third plurality of domains that are known to be benign such that the normalized schema indicates both a plurality of malicious domains and a plurality of benign domains; assigning, to each domain of the plurality of malicious domains, at least one of a tactic or an attribute from a defined list of tactics and attributes; training, based at least in part on the normalized schema, a graph neural network (“GNN”); processing, with the GNN, a first domain that is not included in the normalized schema to determine that the first domain is malicious; determining, for the first domain, at least one of a tactic or an attribute from the defined list of tactics and attributes that provides an indication as to why the first domain is determined to be malicious; and providing an indication that the first domain is malicious and the at least one of the tactic or the attribute determined for the first domain.
2. The computer-implemented method of claim 1, further comprising: generating, for each of a defined plurality of labels, a defined representation; and generating, for each domain of the plurality of domains included in the normalized schema, an embedding, wherein the embedding is generated based at least in part on the defined representations and the domain; and wherein processing the first domain includes: generating, with the GNN, a first embedding for the first domain based at least in part on one or more of the embeddings generated for the plurality of domains included in the normalized schema; and determining, based at least in part on the first embedding, that the first domain is malicious.
3. The computer-implemented method of claim 1, further comprising: defining a graph that includes a plurality of nodes and a plurality of edges that interconnect nodes of the plurality of nodes, wherein: at least some of the nodes of the plurality of nodes represent at least one of a compute instances (“CI”), a domain, an Internet Protocol (“IP”) address, or an account identifier corresponding to the CI; and the GNN is trained based at least in part on the graph.
4. The computer-implemented method of claim 3, wherein at least some of the edges of the plurality of edges are defined as a first interconnection between the domain and the CI, a second interconnection between the domain and the IP address, a third interconnection between the domain and a corresponding second level domain and top-level domain (“TLD”) pair, a fourth interconnection between the domain and a Canonical Name Records (“CNAME”) record, or a fifth interconnection between the domain and the account identifier corresponding to the CI.
5. The computer-implemented method of claim 1, further comprising: monitoring a plurality of compute instances; determining that a first compute instance of the plurality of compute instances is attempting to access the first domain; and in response to determining that the first domain is malicious, blocking access to the first domain by the first compute instance.
6. A computing system, comprising: one or more processors; and a memory storing program instructions that, when executed by the one or more processors, cause the one or more processors to at least: maintain a graph representative of a plurality of domains, wherein: a first sub-plurality of domains of the plurality of domains are known to be benign domains; a second sub-plurality of domains of the plurality of domains are known to be malicious; and a first domain of the plurality is unknown as to whether the first domain is malicious or benign; process, with a graph neural

network (“GNN”), the graph to determine that the first domain is malicious; and in response to determination that the first domain is malicious: determine, for the first domain, at least one of a tactic, an attribute, or a classification that indicates why the first domain is malicious; and block an access with the first domain.

7. The computing system of claim 6, wherein the tactic, the attribute, or the classification is determined from a defined list of tactics, attributes, and classifications.

8. The computing system of claim 7, wherein the at least one of the tactic, the attribute, or the classification is represented by a defined feature that is included in an embedding generated by the GNN for the first domain.

9. The computing system of claim 6, wherein the program instructions that, when executed by the one or more processors to process the graph to determine that the first domain is malicious, further include instructions that, when executed by the one or more processors, further cause the one or more processors to at least: perform a message passing among nodes of the plurality of domains to determine an embedding for the first domain; and determine, based at least in part on the embedding, that the first domain is malicious.

10. The computing system of claim 6, wherein the graph further includes, for each domain of the second sub-plurality of domains, at least one of the tactic or the attribute.

11. The computing system of claim 10, wherein the program instructions that, when executed by the one or more processors, further cause the one or more processors to at least: periodically update and process the graph to determine, for each domain of the plurality of domains, whether the domain is benign or malicious.

12. The computing system of claim 6, wherein: the graph further includes, for each domain of the plurality of domains, an embedding representative of the domain; and features included in each embedding are selected from a defined set of features.

13. The computing system of claim 6, wherein: the graph includes a plurality of nodes and a plurality of edges; at least some of the nodes of the plurality of nodes represent at least one of a compute instances (“CI”), a domain of the plurality of domains, an Internet Protocol (“IP”) address, or an account identifier corresponding to the CI; and at least some of the edges of the plurality of edges are defined as a first interconnection between the domain and the CI, a second interconnection between the domain and the IP address, a third interconnection between the domain and a corresponding second level domain and top-level domain (“TLD”) pair, a fourth interconnection between the domain and a Canonical Name Records (“CNAME”) record, or a fifth interconnection between the domain and the account identifier corresponding to the CI.

14. The computing system of claim 6, wherein the program instructions that, when executed by the one or more processors, further cause the one or more processors to at least: monitor a compute instance (“CI”) to detect an attempted access to the first domain; in response to the attempted access, process the graph to determine that the first domain is malicious; and block the access between the CI and the first domain.

15. The computing system of claim 6, wherein the program instructions that, when executed by the one or more processors, further cause the one or more processors to at least: in response to determination that the first domain is malicious, send an indication as to why the first domain is determined to be malicious.

16. A method, comprising: defining a first node of a graph as a first embedding representative of a benign domain; defining a second node of the graph as a second embedding representative of a malicious domain; defining a third node of the graph as corresponding to a first domain that is unknown as to whether the first domain is a malicious domain or a benign domain; processing the graph with a graph neural network to determine a third embedding for the third node, wherein the third embedding is determined based at least in part on the first embedding, the second embedding, and at least one edge that interconnects the third node to a fourth node that is also connected to at least one of the first node and the second node; determining, based at least in part on the third

embedding, that the first domain is malicious and at least one of a tactic or an attribute determined for the first domain; and in response to determining that the first domain is malicious, blocking an access with the first domain.

17. The method of claim 16, further comprising: generating the first embedding based at least in part on features defined for a plurality of labels.

18. The method of claim 16, further comprising: generating the second embedding to include a representation of at least one of a second tactic or a second attribute that indicates why the first domain is malicious.

19. The method of claim 16, further comprising: in response to determining that the first domain is malicious, generating an indication as to why the first domain is malicious.

20. The method of claim 16, further comprising: defining the second embedding based on one or more of a matrix decomposition over an adjacency matrix, an adjacency-based similarity embedding, or a transformation of label correlations into a knowledge graph with a plurality of relations.
